KLH
– BACHUPALLY –

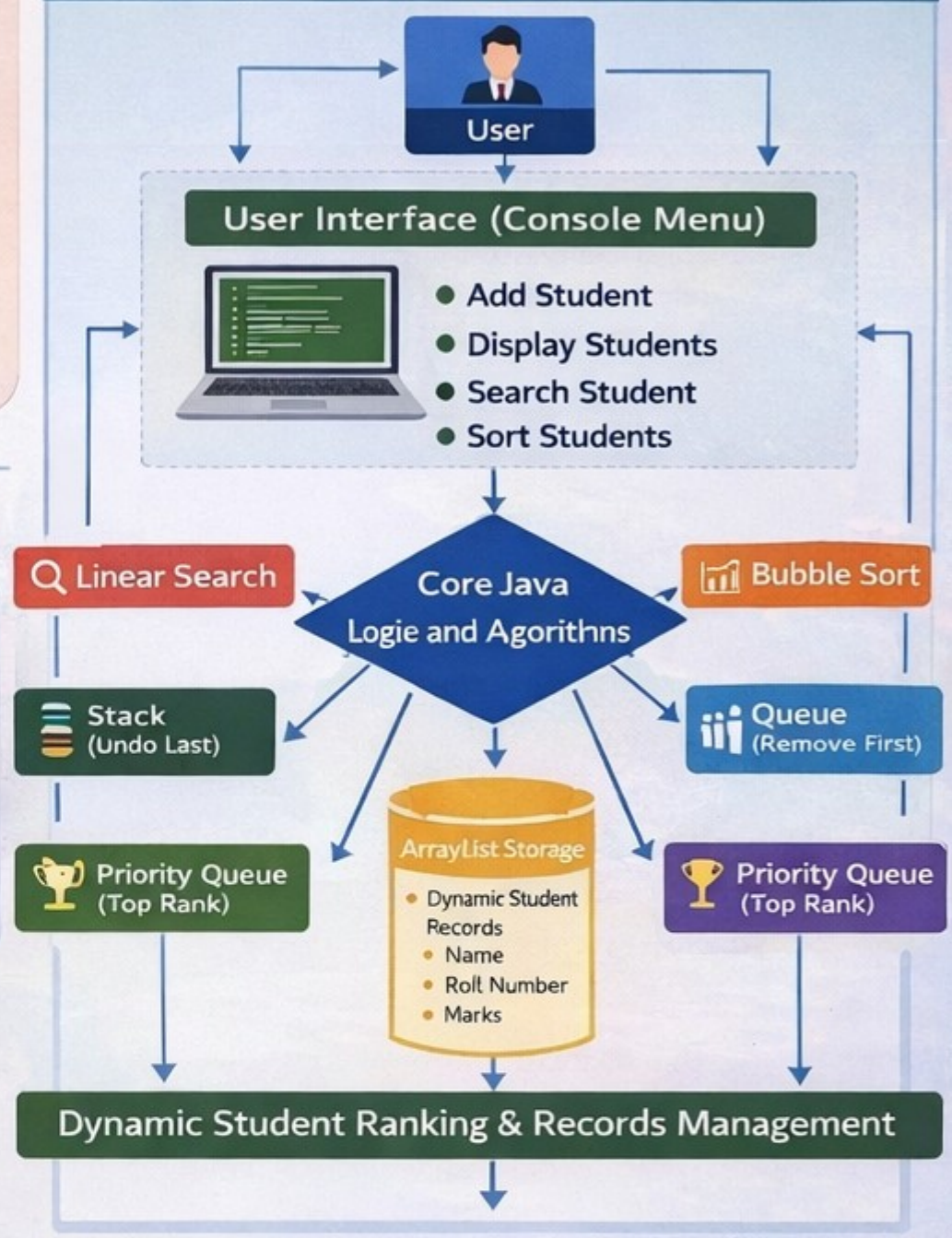
DYNAMIC STUDENT RANKING AND RECORDS

PROJECT BASED LEARNING – DATA STRUCTURES AND ALGORITHMS

ABSTRACT

The Dynamic Student Ranking and Records System is a Java-based application that manages student records using Data Structures like Array-List, Stack, Queue, and Priority Queue. It implem-eents algorithms such as **Linear** Search and Bubble Sort to efficiently store, search, sort, and rank student data.

SYSTEM ARCHITECTURE



CO-WISE IMPLEMENTATION

CO1 Searching and Sorting Algorithms

- ✔ Linear Search implementation to find student
- ✔ Bubble Sort used to rank students by marks
- ✔ Comparison and iteration for algorithm execut-tion, Efficient organization of student data

CO1

```
==== Dynamic Student Ranking And Records ==
1. Add Student
2. Display Students
3. Search Student
4. Sort Students By Marks
5 Stack Pop (Undo Last Added)
6 Queue Remove (First Added)
7. Exit
Enter Choice:
```

CO3 Stack, Queue, Priority Queue

- ✔ Stack used to undo last added student record
- ✔ Queue used to remove first inserted student
- ✔ Priority Queue used (for top rank student
- ✔ Demonstrates. LiFO, FIFO and priority processing

CO3

CO2 List ABSTRACT DATA TYPE (ADT)

- ✔ Implementation of ArrayList to store student records
- ✔ Dynamic insertion and deletion of student data
- ✔ Traversing list to display student etails
- ✔ Managing records using, Java Collection List

CO4 Java Collections Framework

- ✔ Use of ArrayList, Stack, Queue, PriorityQueue
- ✔ Efficient storage and retrieval of data
- ✔ Built-in Java data structures for better performance
- ✔ Simplifies implementation of complex operations

Name: G. PAVANI
Roll No: 2520030046
Cluster-2, Sec-6